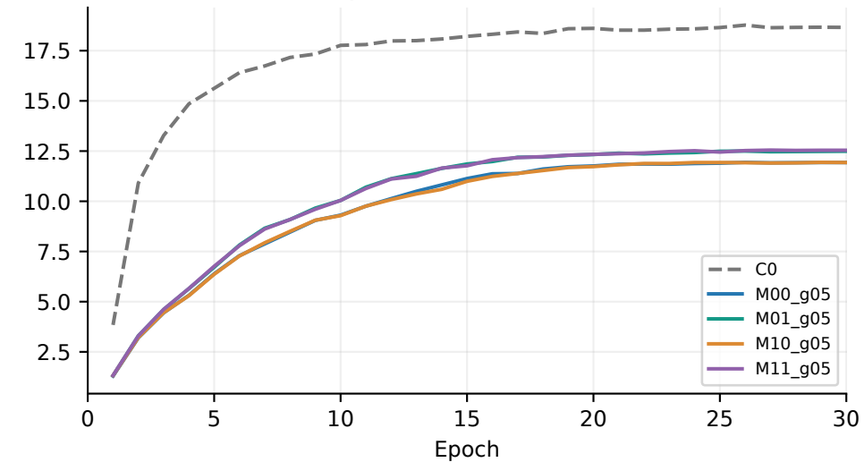
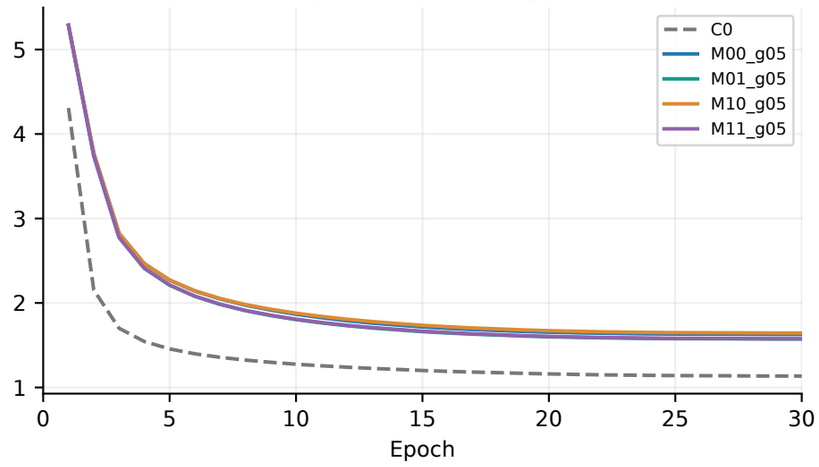
512D VAST frozen-head training | seed 50 | completed epochs only

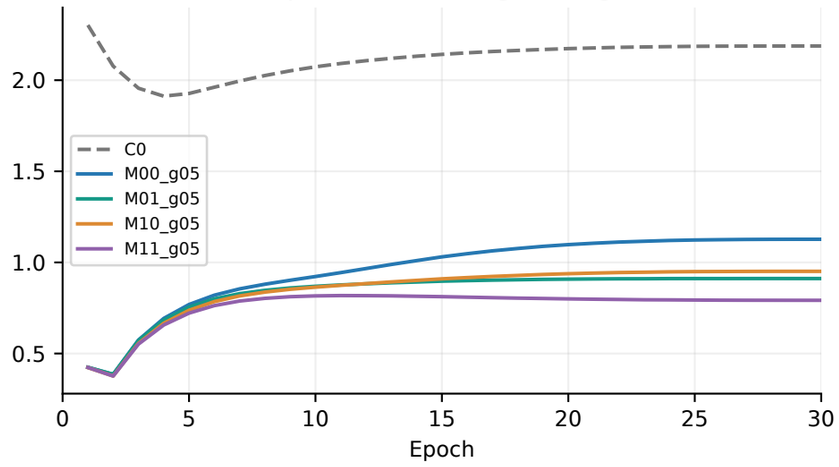
Initial scale multiplier 0.5: Validation macro R@1 (%)



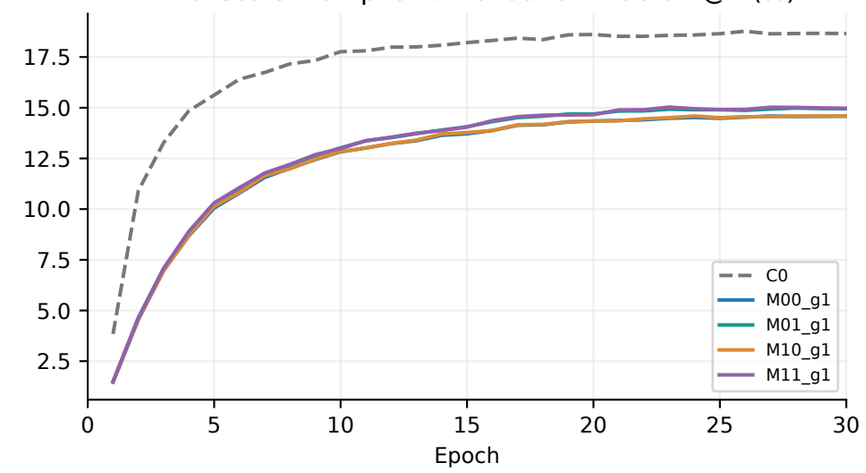
Initial scale multiplier 0.5: Training contrastive loss



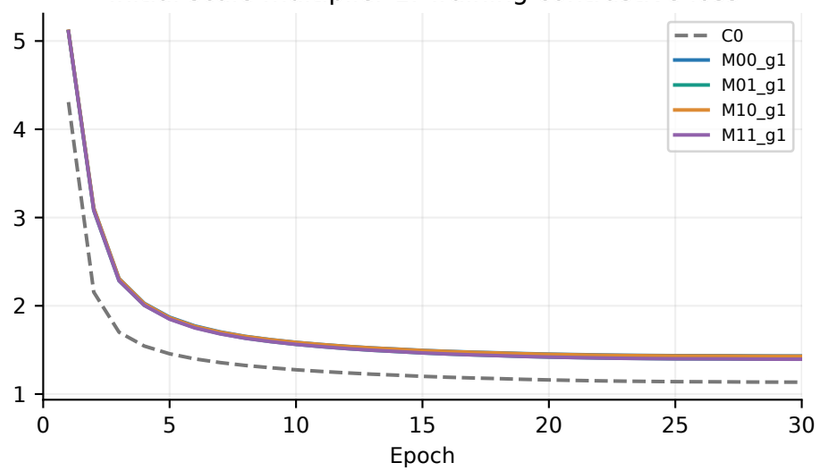
Initial scale multiplier 0.5: Training unweighted entailment



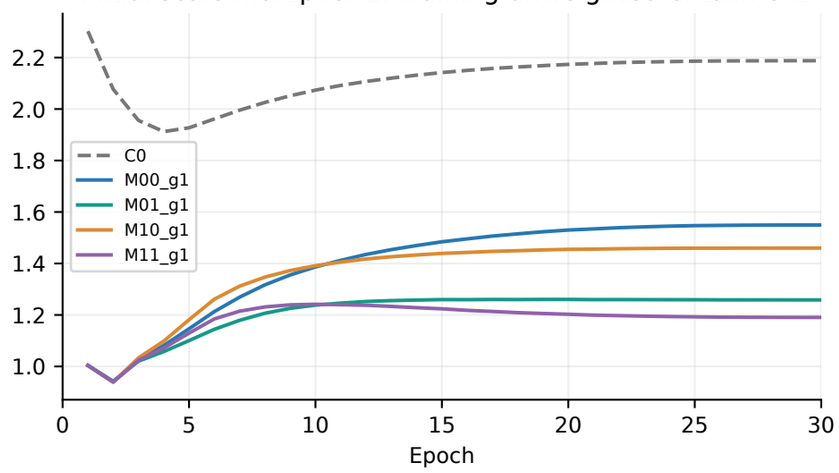
Initial scale multiplier 1: Validation macro R@1 (%)



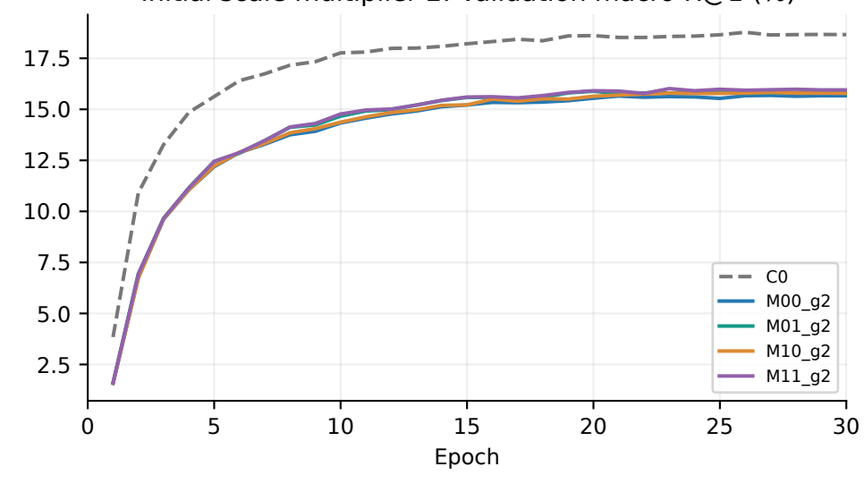
Initial scale multiplier 1: Training contrastive loss



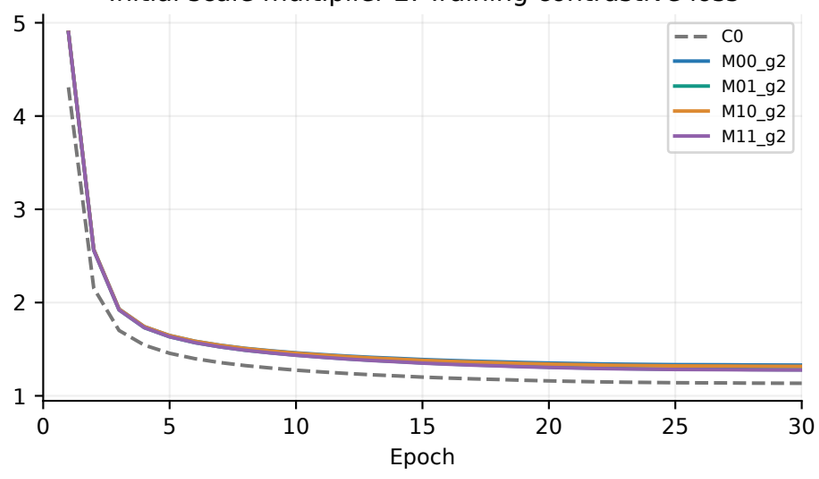
Initial scale multiplier 1: Training unweighted entailment



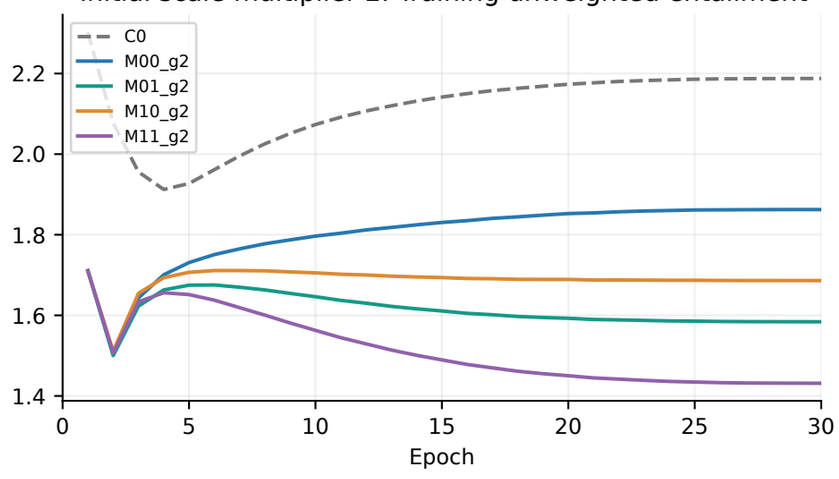
Initial scale multiplier 2: Validation macro R@1 (%)



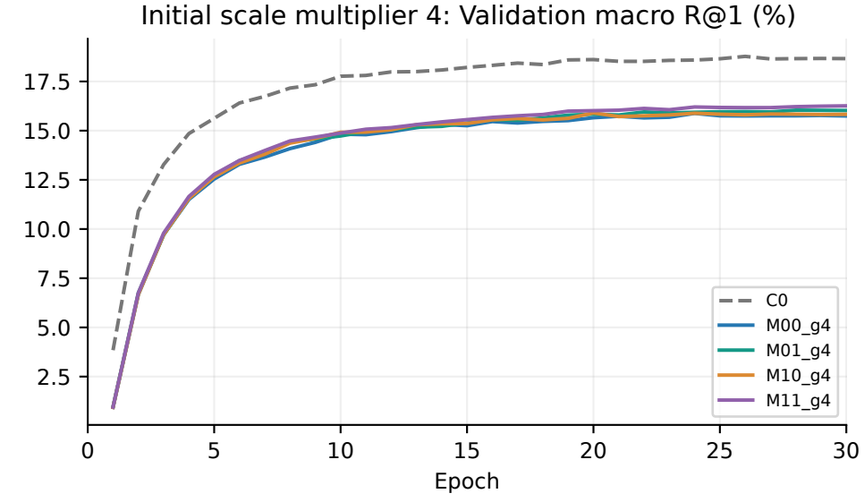
Initial scale multiplier 2: Training contrastive loss



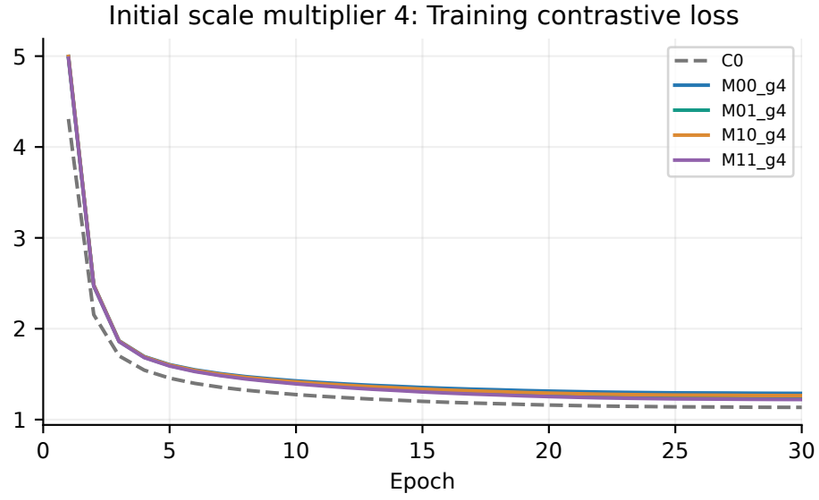
Initial scale multiplier 2: Training unweighted entailment



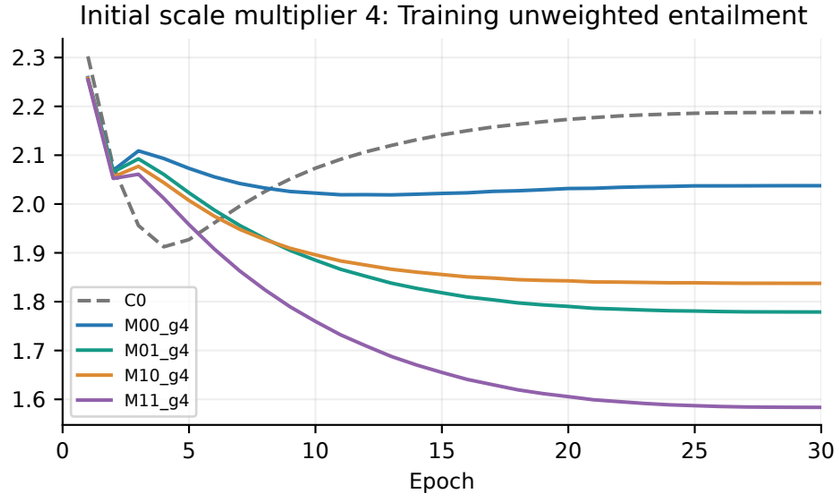
Initial scale multiplier 4: Validation macro R@1 (%)



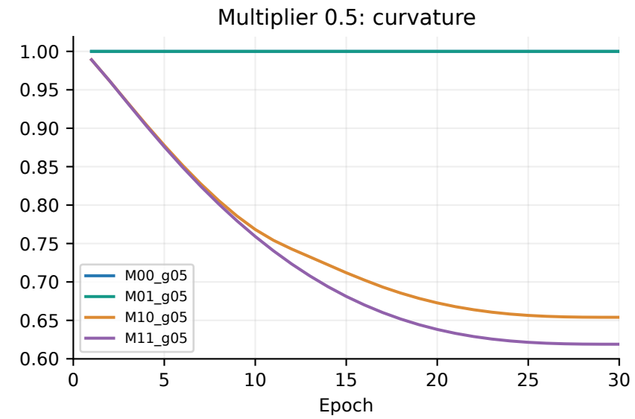
Initial scale multiplier 4: Training contrastive loss

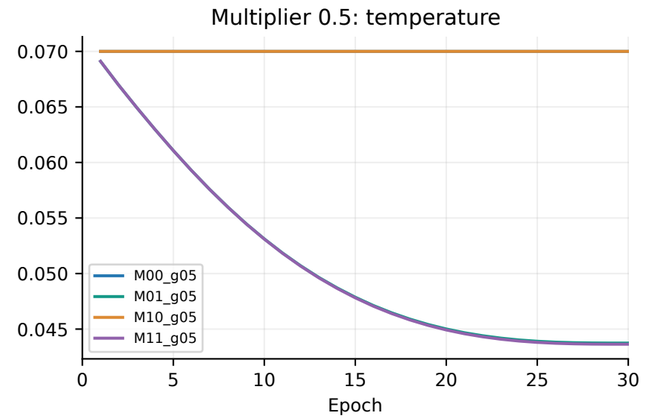


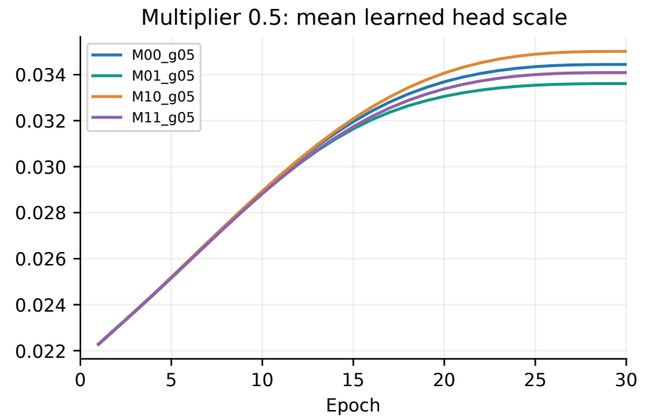
Initial scale multiplier 4: Training unweighted entailment

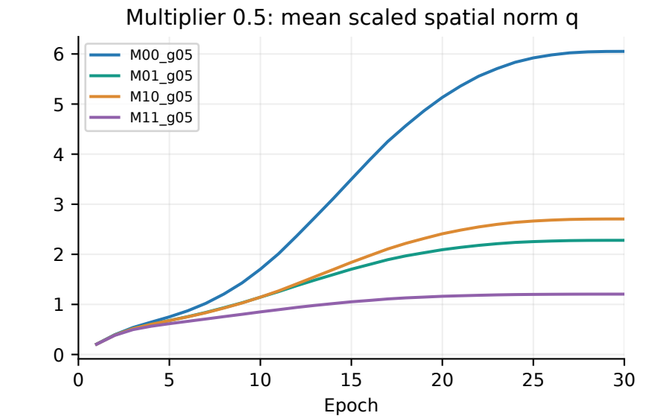


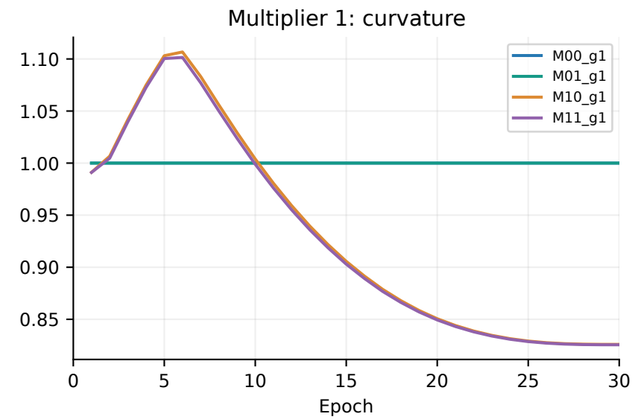
Learned scalar trajectories (separate from geometric success)

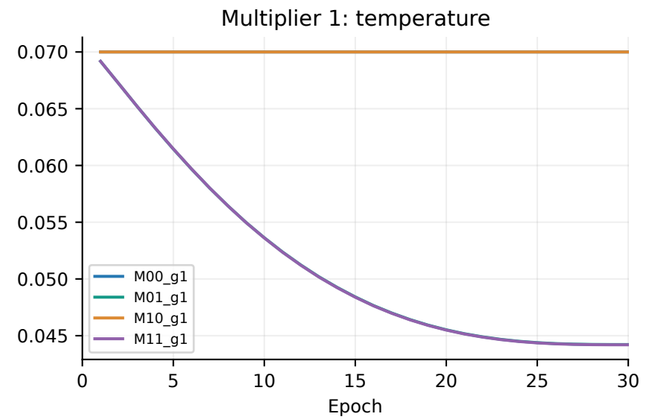


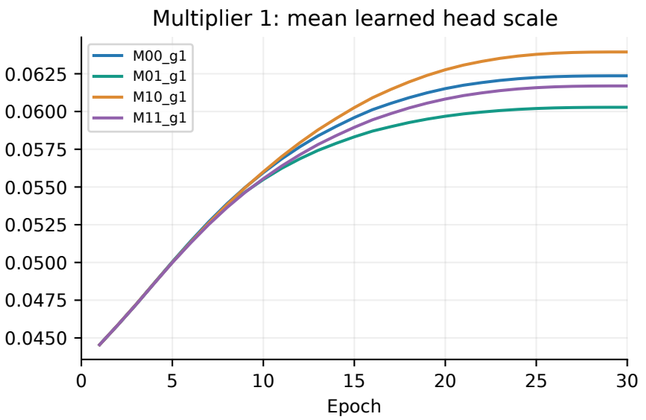


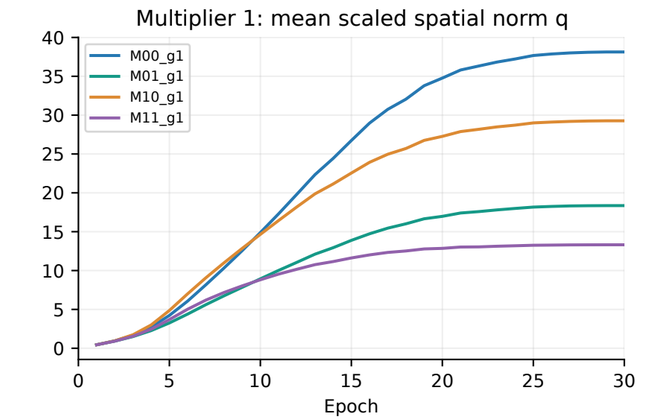


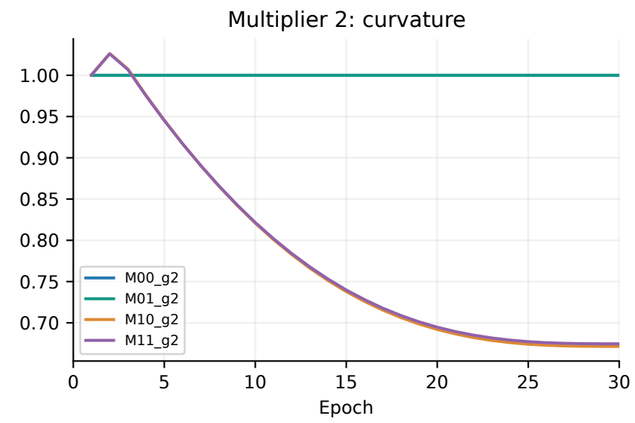


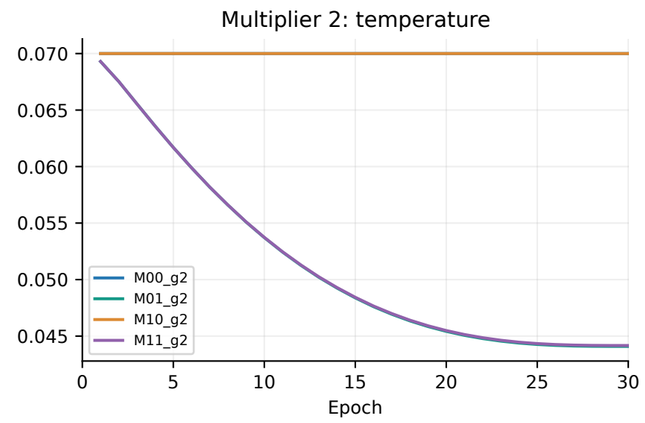


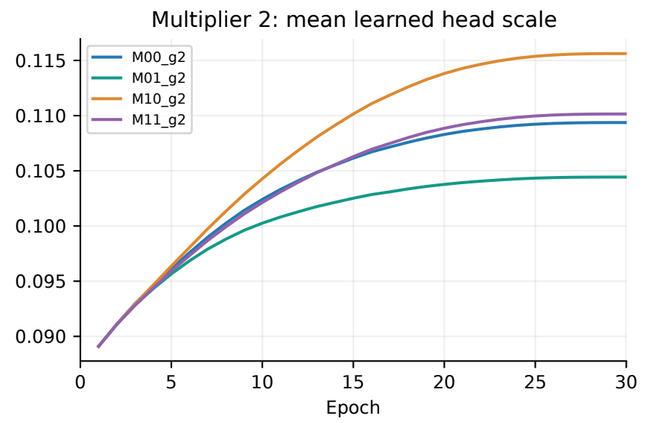


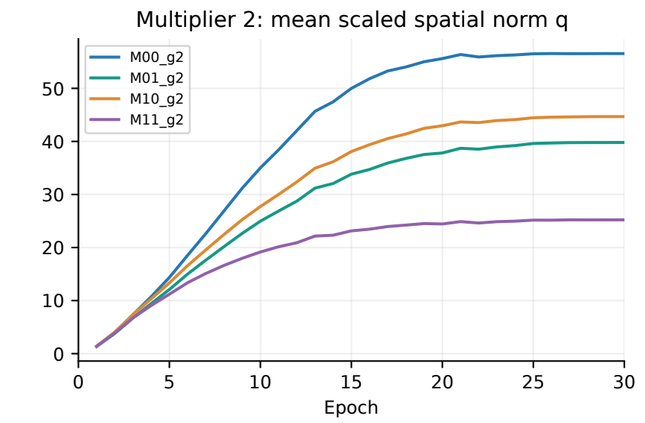


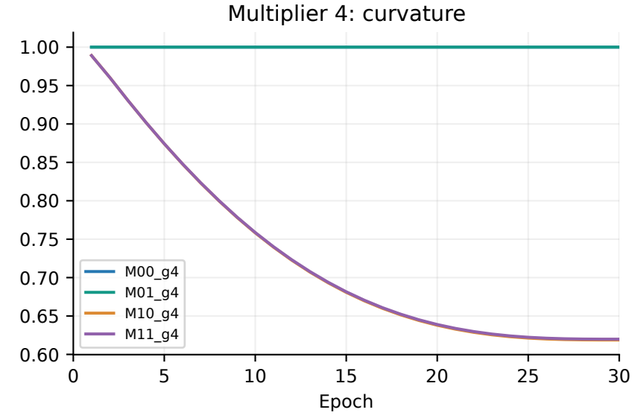


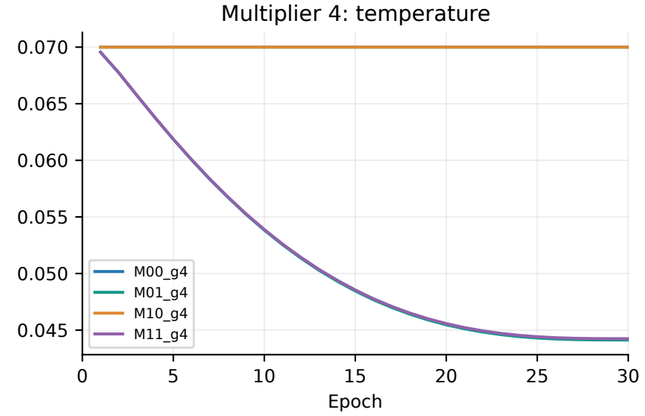


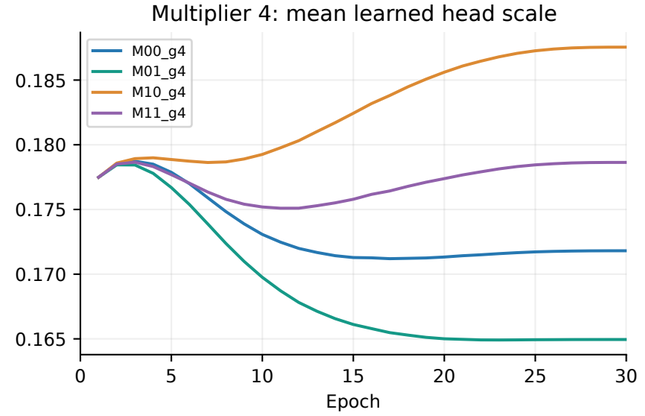


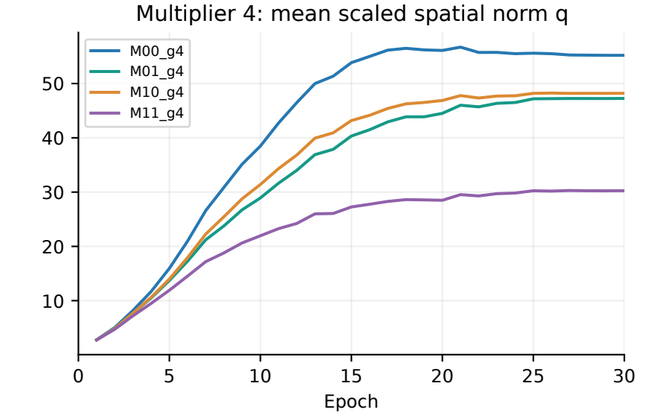


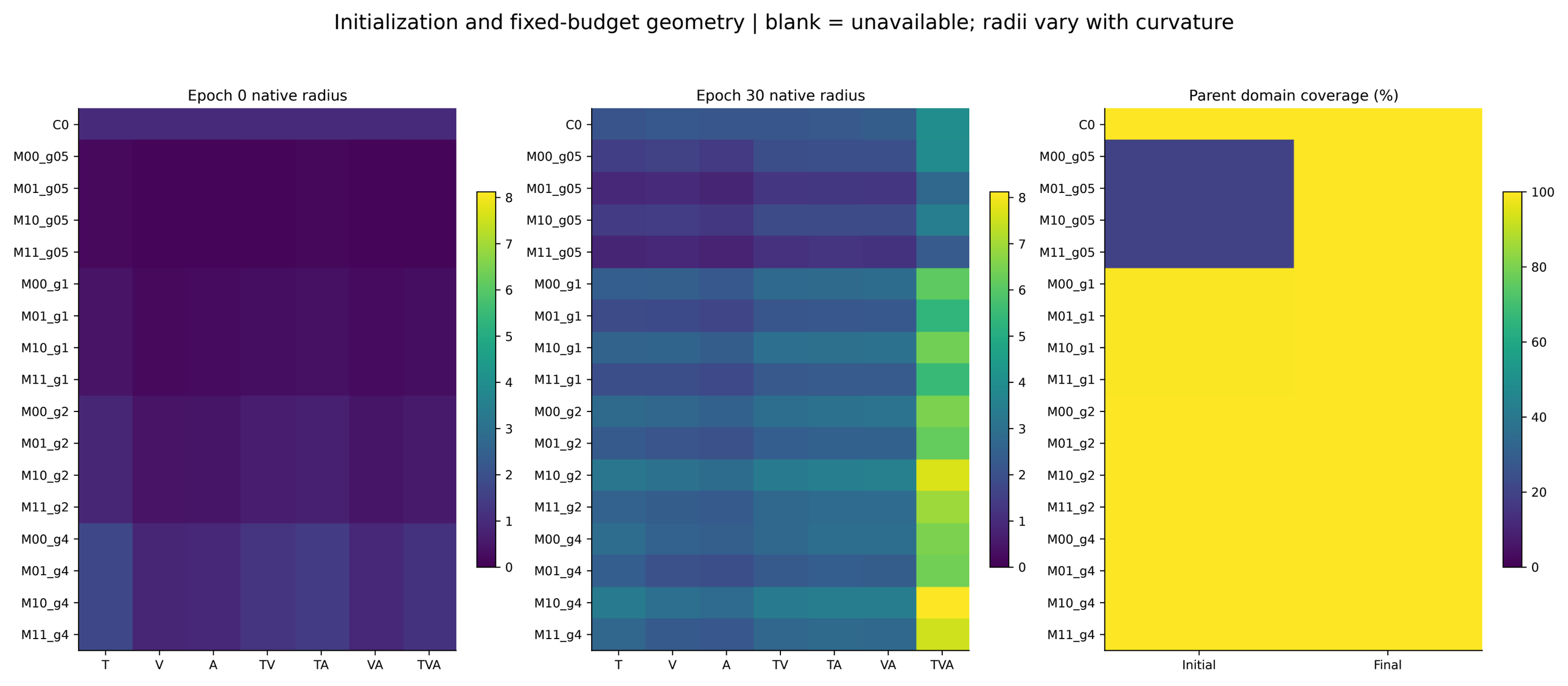


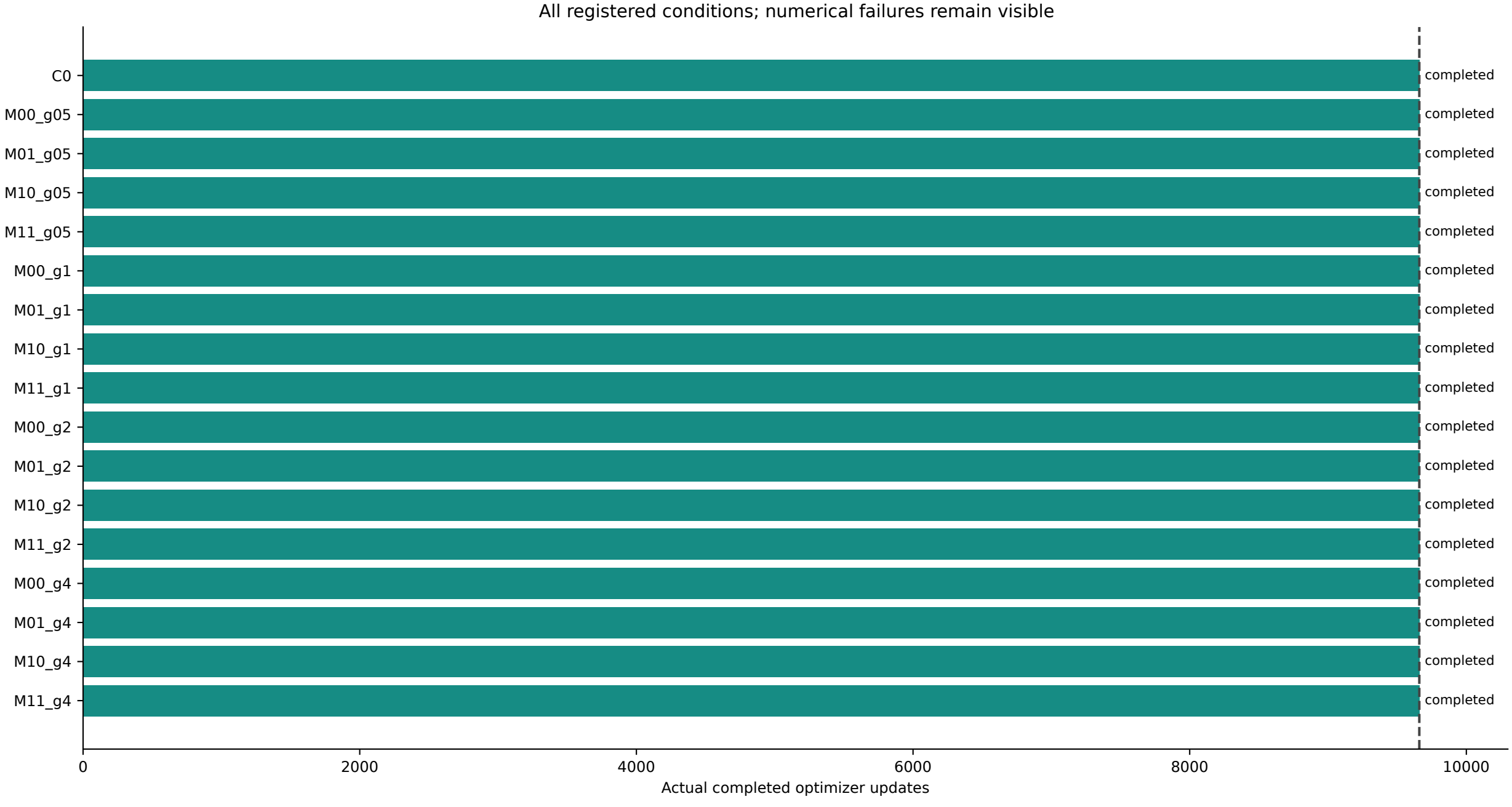




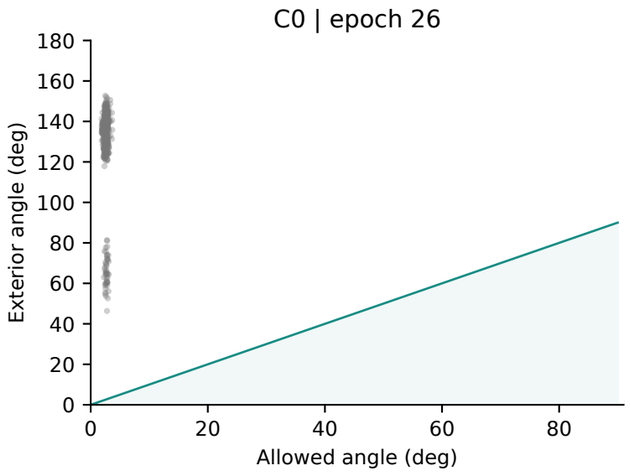


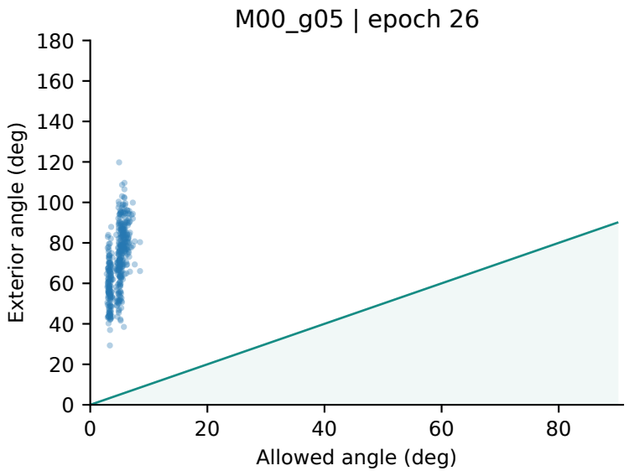


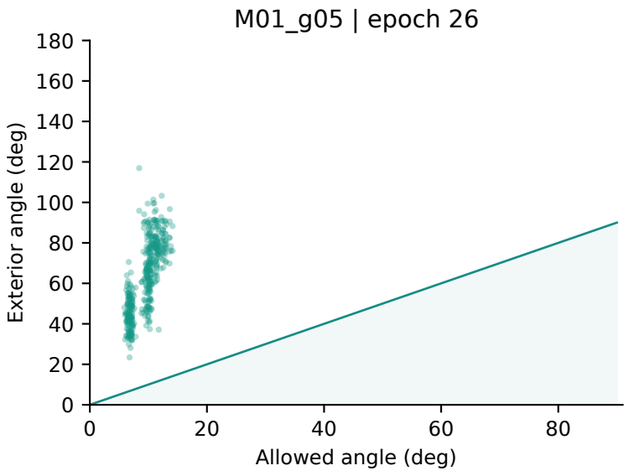

Initialization and fixed-budget geometry | blank = unavailable; radii vary with curvature
Epoch 0 native radius
Epoch 30 native radius
Parent domain coverage (%)
C0
M00_g05
M01_g05
M10_g05
M11_g05
M00_g1
M01_g1
M10_g1
M11_g1
M00_g2
M01_g2
M10_g2
M11_g2
M00_g4
M01_g4
M10_g4
M11_g4
T
V
A
TV
TA
VA
TVA
Initial
Final
0
1
2
3
4
5
6
7
8
0
20
40
60
80
100

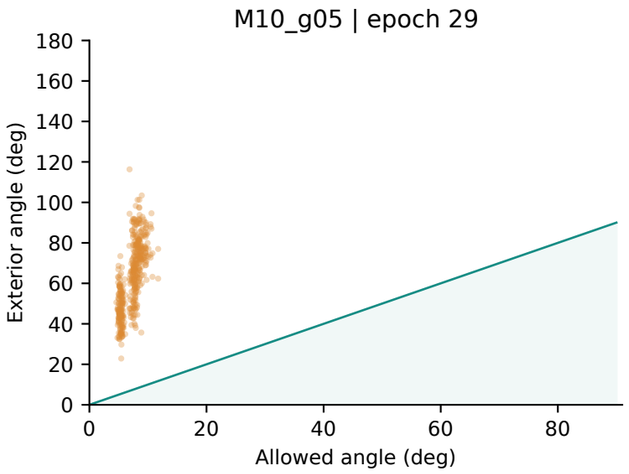

All registered conditions; numerical failures remain visible
C0
M00_g05
M01_g05
M10_g05
M11_g05
M00_g1
M01_g1
M10_g1
M11_g1
M00_g2
M01_g2
M10_g2
M11_g2
M00_g4
M01_g4
M10_g4
M11_g4
completed
0
2000
4000
6000
8000
10000
Actual completed optimizer updates

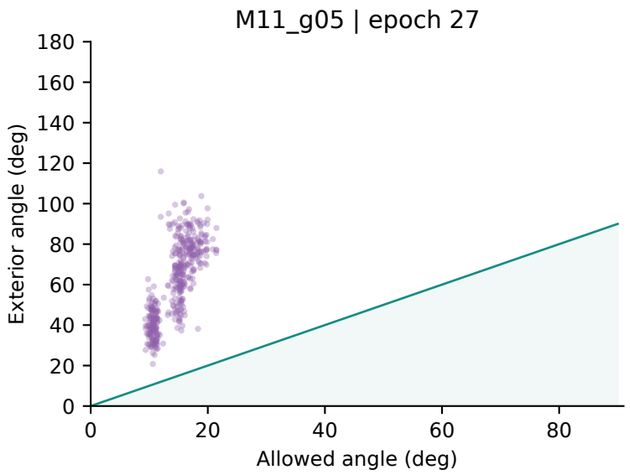
Native FP32 cone angles | same 48 validation videos x 9 edges | best available saved readout

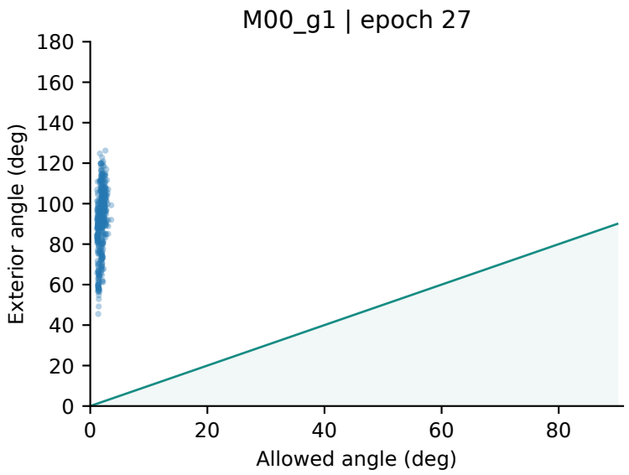


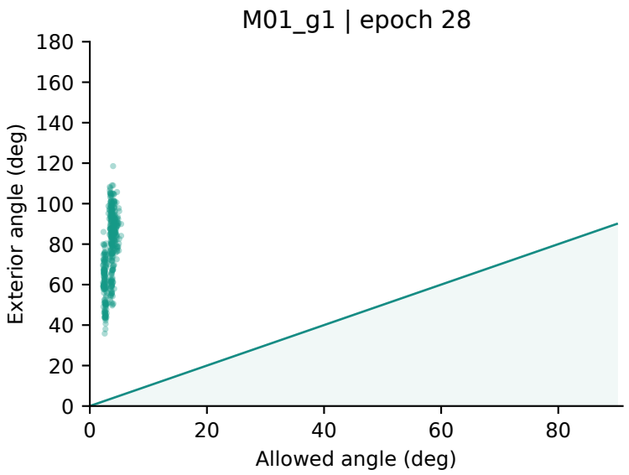


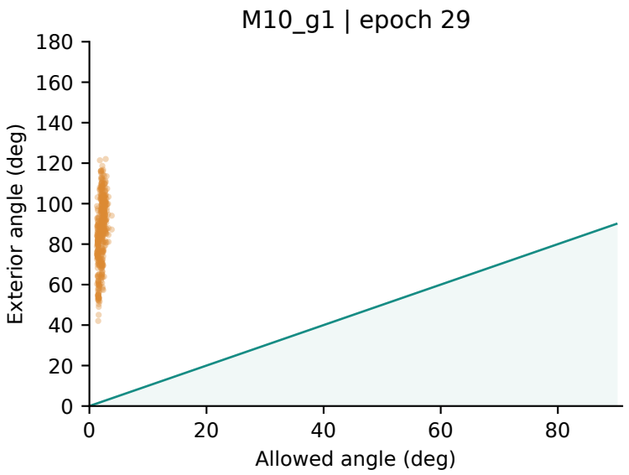


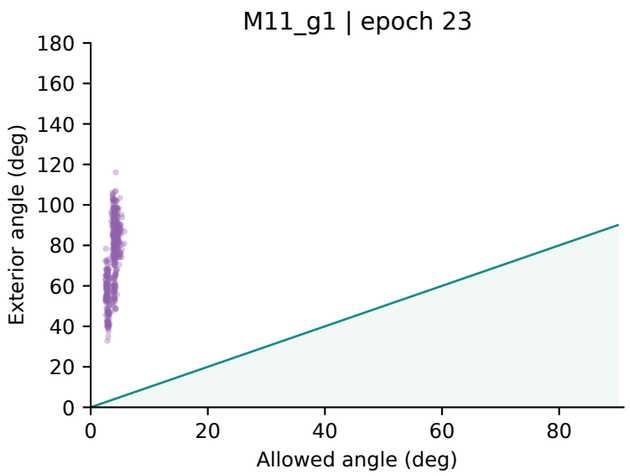


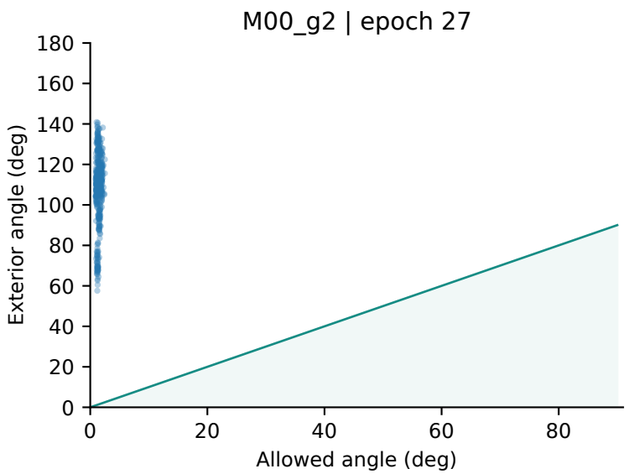


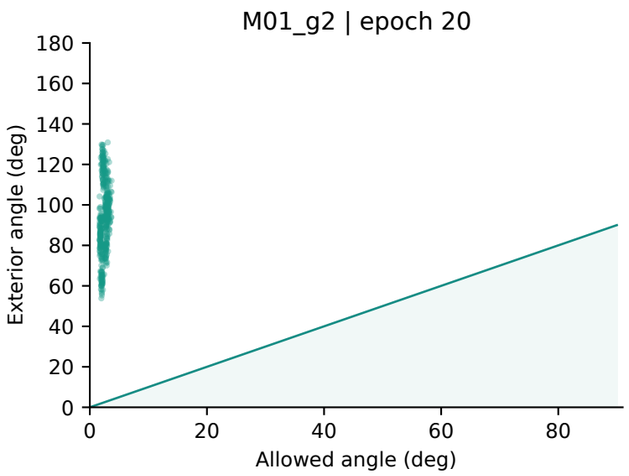


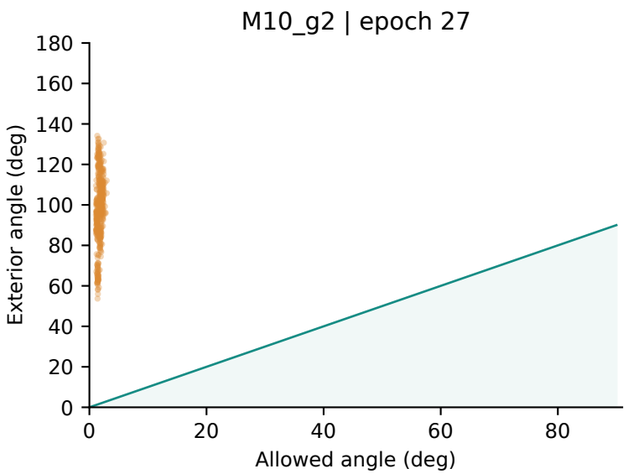


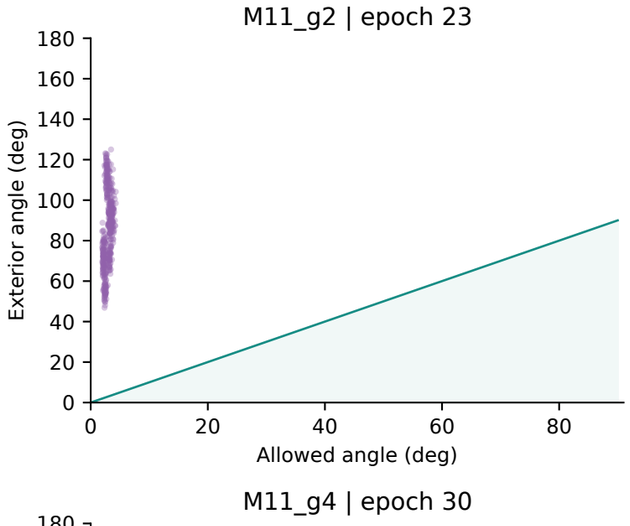


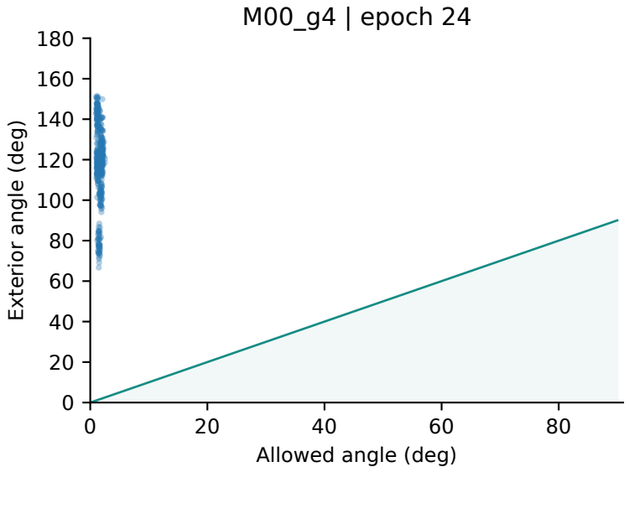


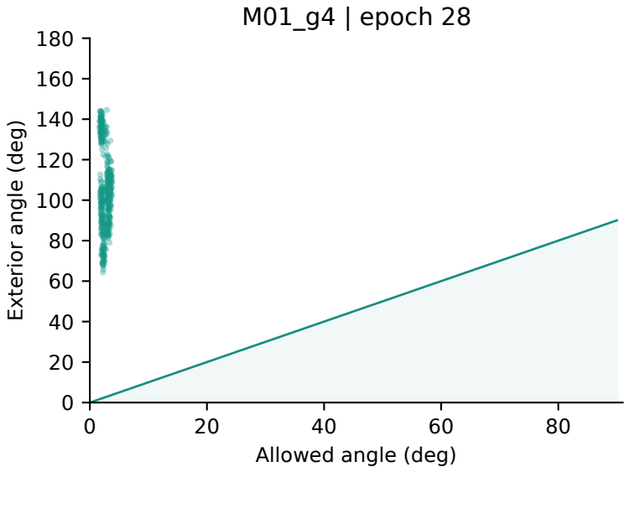


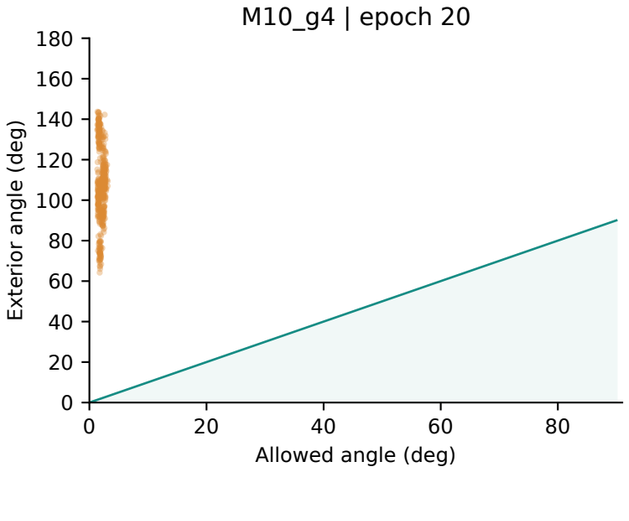


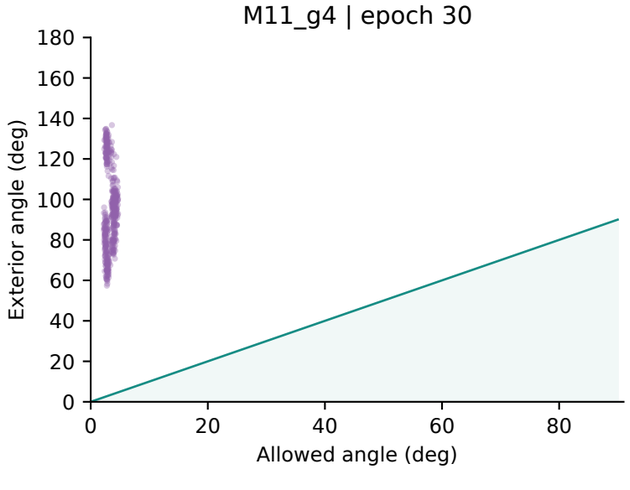






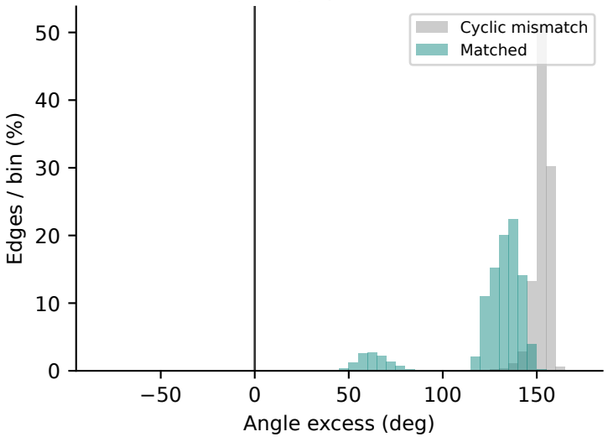




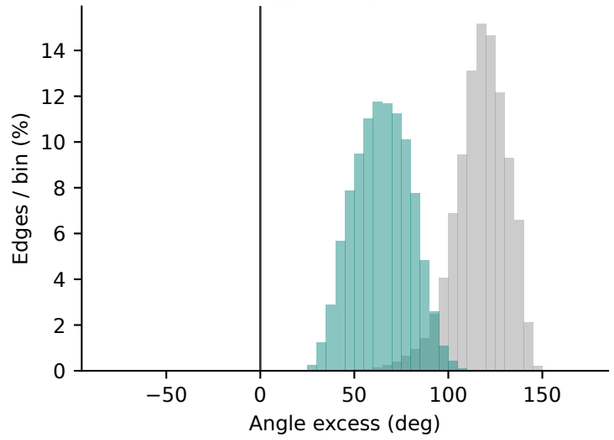


Native FP32 signed violations | all 2048 validation videos | best available saved readout

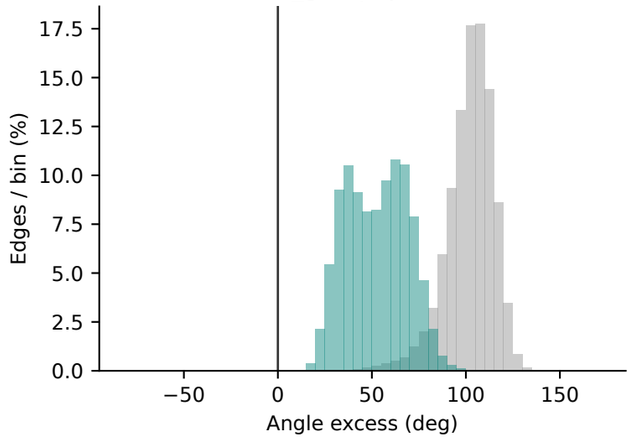
C0 | epoch 26



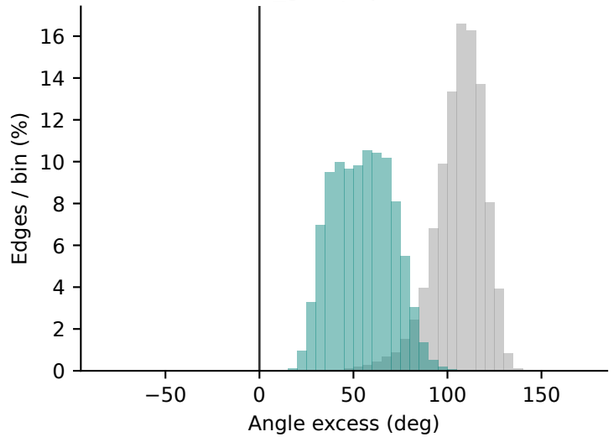
M00_g05 | epoch 26



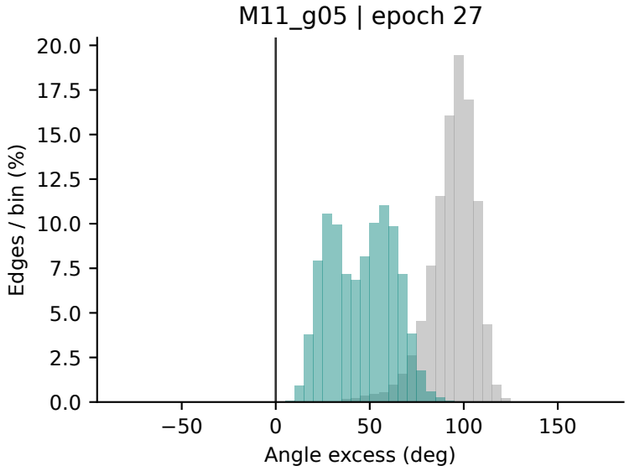
M01_g05 | epoch 26



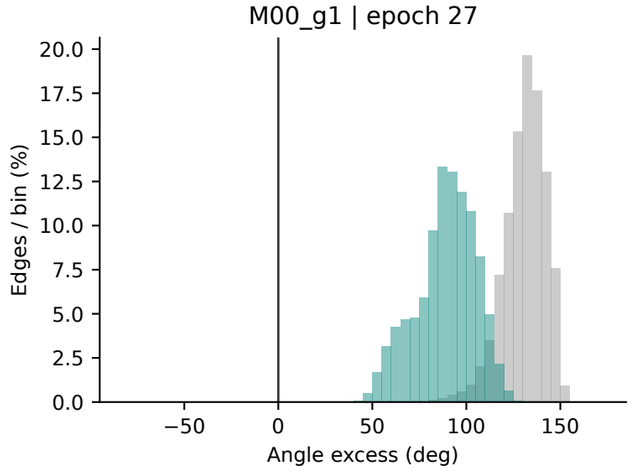
M10_g05 | epoch 29



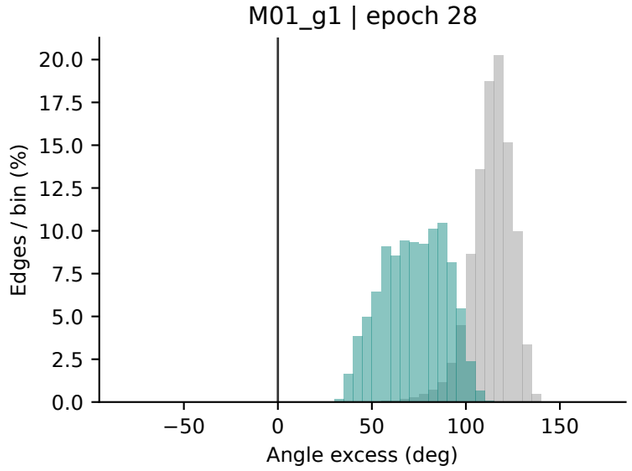
M11_g05 | epoch 27



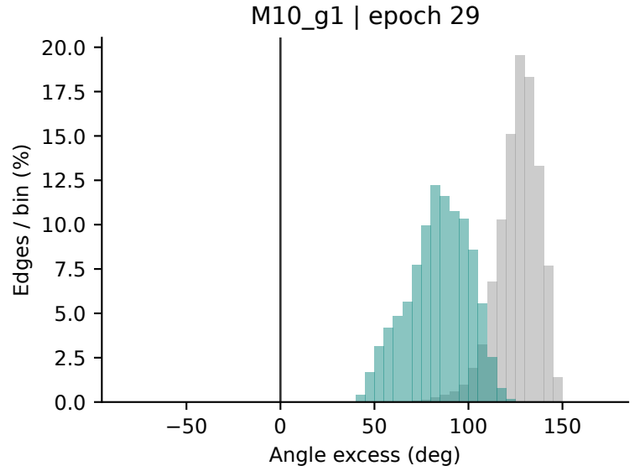
M00_g1 | epoch 27



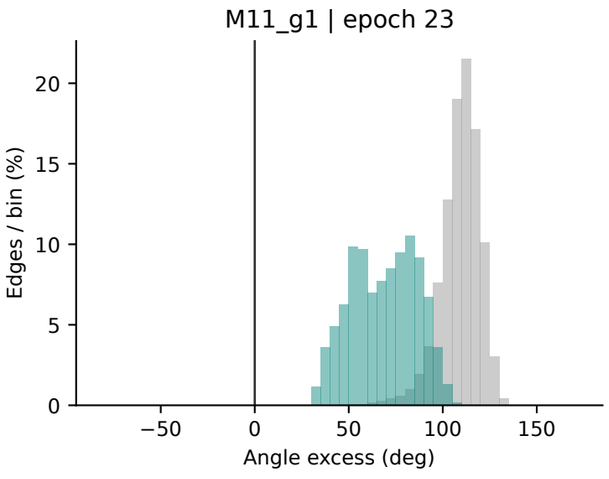
M01_g1 | epoch 28



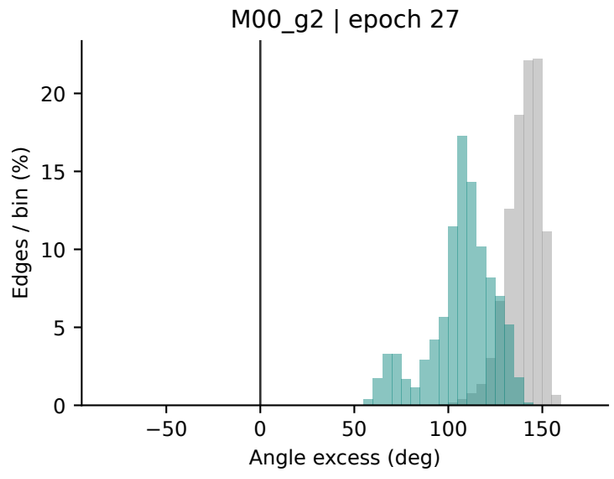
M10_g1 | epoch 29



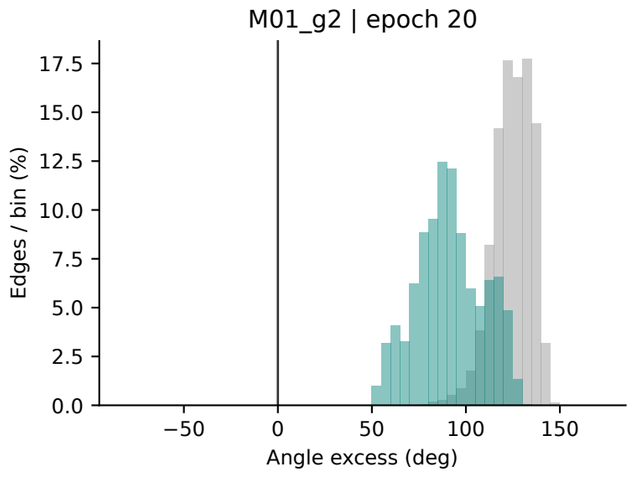
M11_g1 | epoch 23



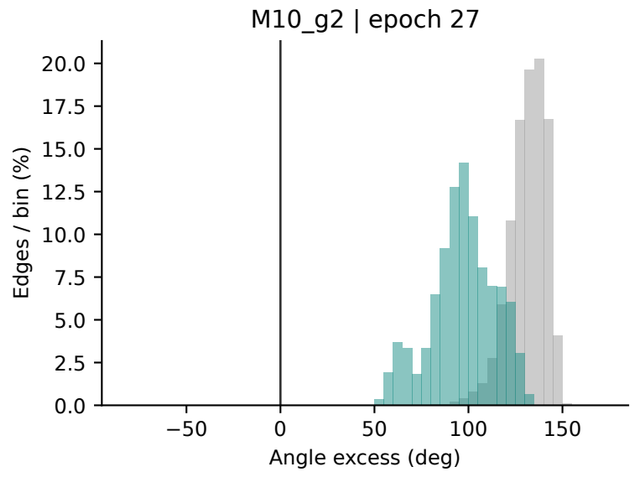
M00_g2 | epoch 27



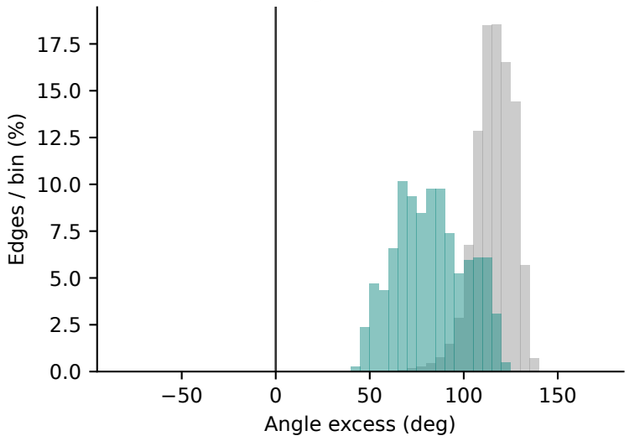
M01_g2 | epoch 20



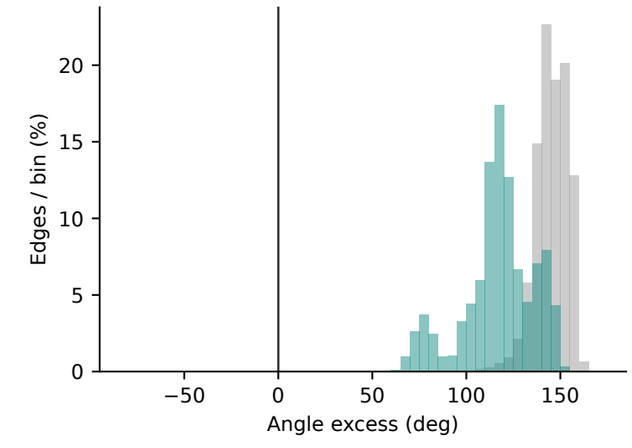
M10_g2 | epoch 27



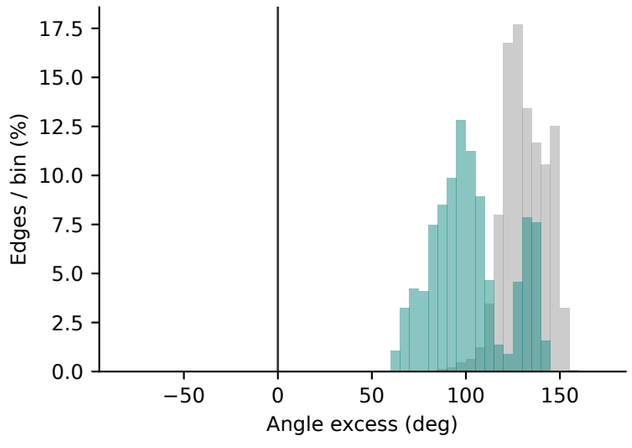
M11_g2 | epoch 23



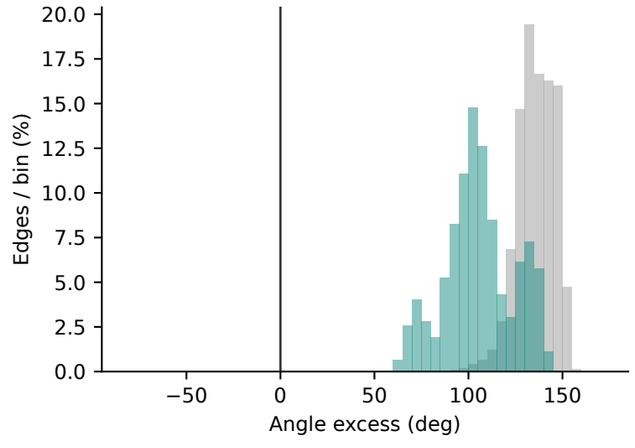
M00_g4 | epoch 24



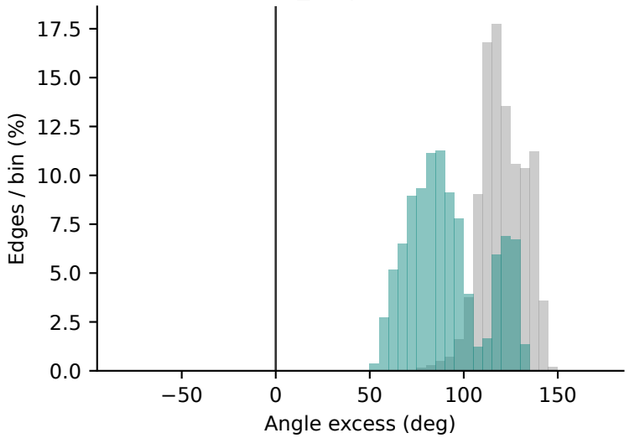
M01_g4 | epoch 28



M10_g4 | epoch 20

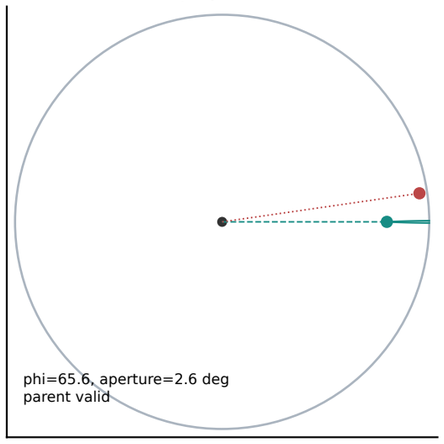


M11_g4 | epoch 30

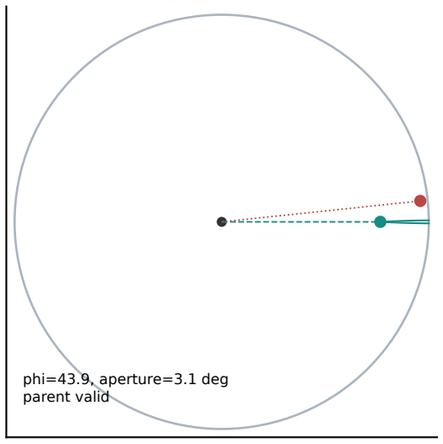


Exact O-TV-TVA planes | same fixed validation index 26 | green TV, red TVA | best available readout

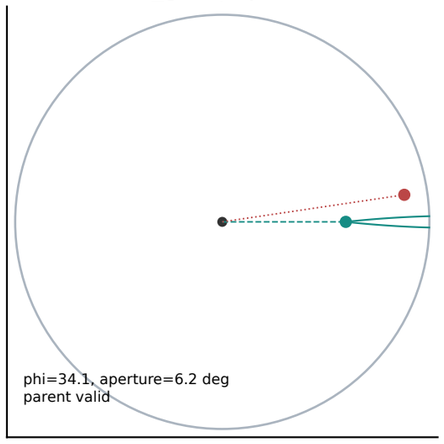
C0 | epoch 26



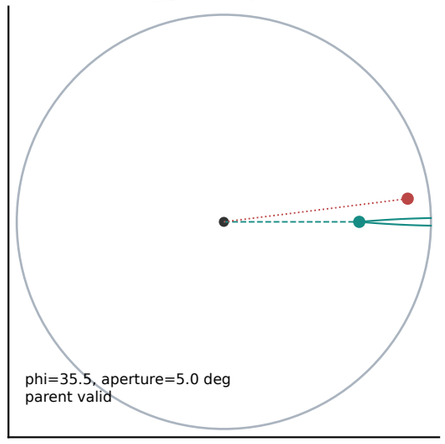
M00_g05 | epoch 26



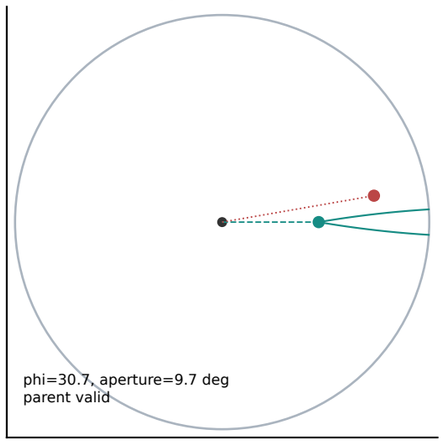
M01_g05 | epoch 26



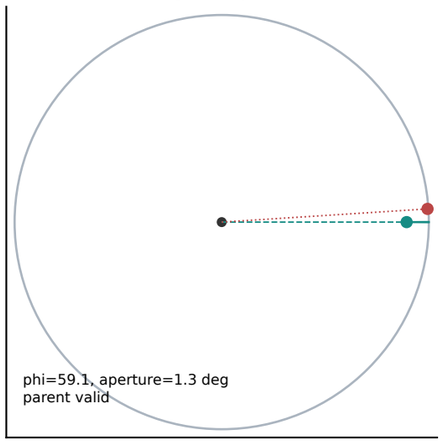
M10_g05 | epoch 29



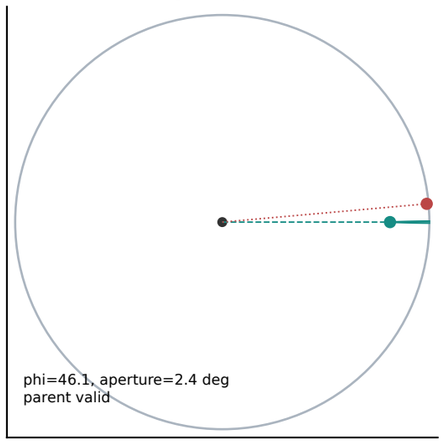
M11_g05 | epoch 27



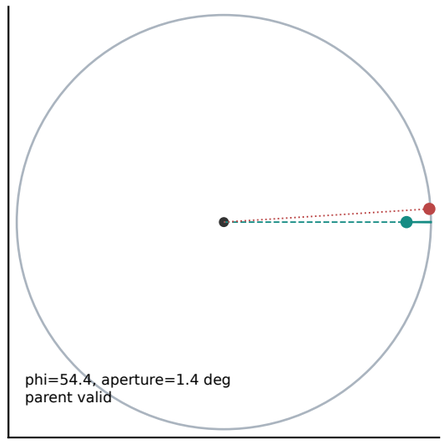
M00_g1 | epoch 27



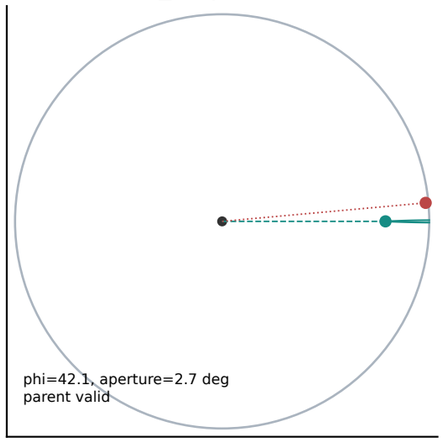
M01_g1 | epoch 28



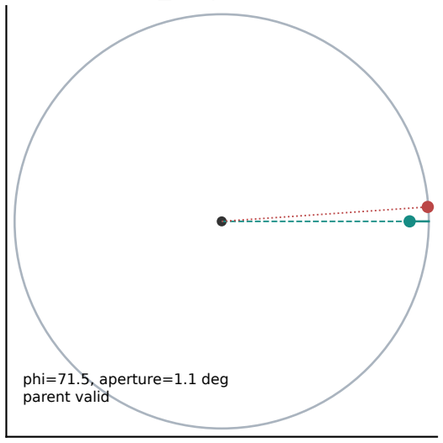
M10_g1 | epoch 29



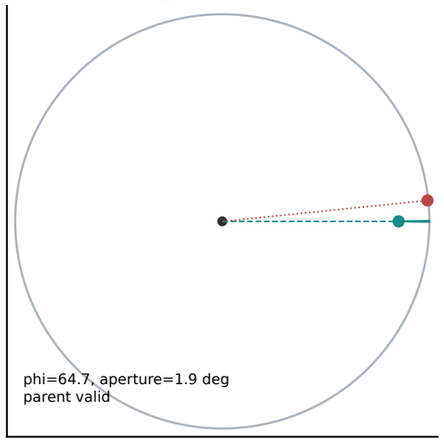
M11_g1 | epoch 23



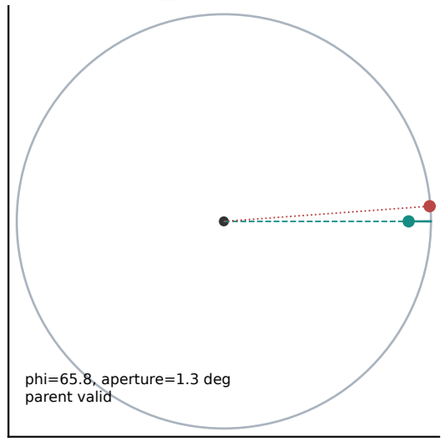
M00_g2 | epoch 27



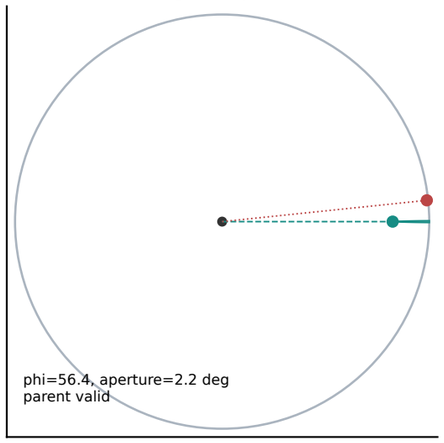
M01_g2 | epoch 20



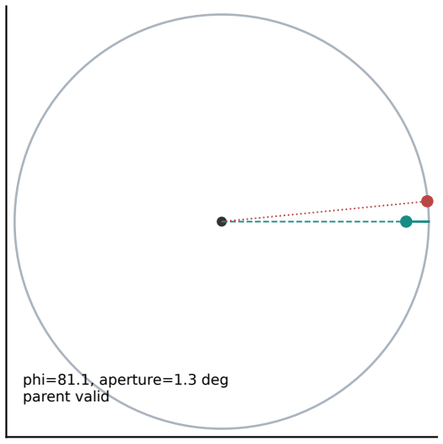
M10_g2 | epoch 27



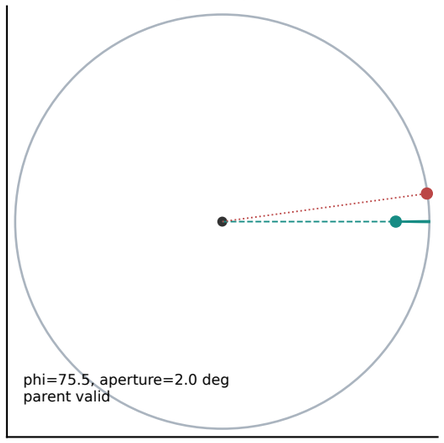
M11_g2 | epoch 23



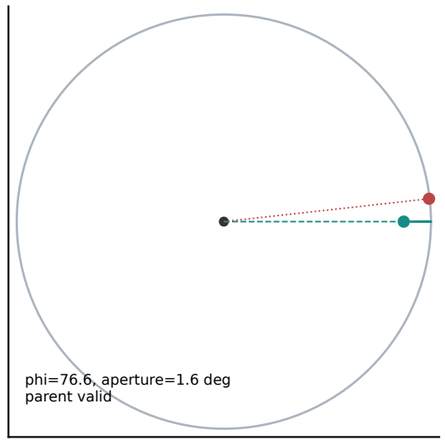
M00_g4 | epoch 24



M01_g4 | epoch 28



M10_g4 | epoch 20



M11_g4 | epoch 30

